KIRANA A P

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Summary

Recent Robotics and Artificial Intelligence graduate with strong fundamentals in networking, Linux, Python, SQL, and embedded systems. Hands-on experience in developing AI and IoT-based projects, with strong problem-solving, troubleshooting, and teamwork skills. Eager to learn new technologies and contribute to innovative IT solutions.

Skills

Networking: TCP/IP, OSI Model, IPv4, Subnetting, VLAN, Routing, Switching, DHCP, DNS
Programming Languages: Python, SQL, C++
Robotics & Designs: ROS2, Arduino, ESP32, LLM, AutoCAD, Fusion360, Solid Edge
Database: SQL, POWER BI, EXCEL
Tools: Git& GitHub, Docker, Jenkins, YAML
Operating Systems: Linux (Ubuntu), Windows

Experience

iSpark Learning Solution **February 2026**

Tools Used: Python, ROS2

- **Architechted** a **ROS2-inspired** autonomous robotics platform leveraging AI vision, embedded control, MQTT communication, and autonomous navigation, reducing manual intervention by 80%.
- **Designed** perception and navigation modules using **OpenCV, Raspberry Pi, and sensor fusion** techniques, improving obstacle-detection reliability to **95%.**

Projects

Enterprise Network Design | *Cisco Packet Tracer*

- Developed hands-on networking skills through **Cisco Packet Tracer** by configuring enterprise network topologies with **VLANs, trunk ports, and router-based inter-VLAN routing.**
- Proficient in **Cisco IOS command-line interface (CLI), including switch and router configuration, VLAN management, and basic network troubleshooting.**

Autonomous Hexapod Robot with SLM Integration | *PyTorch, OpenCV*

- **Engineered** a 6-legged autonomous robot using **ESP32, Arduino Uno, and 18 servo motors**, implementing tripod-gait locomotion and **IMU-based stabilization** to improve terrain adaptability and movement accuracy.
- **Developed** a multi-modal AI perception system integrating **YOLOv5s, SLM, ToF, and gas sensors**, generating real-time environmental summaries through a Wi-Fi dashboard for rescue and exploration scenarios.

AI-Powered Natural Language SQL Dashboard (SaaS) | *Python, SQLAlchemy*

- **Built** a Natural Language-to-SQL analytics platform using **Python, Streamlit, OpenAI API, SQLAlchemy, and Pandas,** reducing manual query-writing effort by **90%** and enabling self-service business intelligence.
- **Automated** data ingestion and interactive reporting workflows for **CSV, SQLite, and MySQL** sources, cutting dataset onboarding time by **87%** and report generation time by **70%.**

Awards & Certifications

- **National Level Robothon (eYantra)-2025:** Participated and Built autonomous a Grid Line Following Robot.
- **Best Final Year Project – 2026, BIT College:** Awarded 3rd Place in Robo Cell Competition (BIT, 2025) for developing an AI-powered autonomous hexapod robot with real-time data processing

Education

B.E. Robotics and AI **Graduated: 2026**
Bangalore Institute Of Technology *CGPA: 8 / 10*

Diploma. Mechanical Engineering **Completed: 2023**
Sahyadri Polytechnic, Thirthahalli *CGPA: 8.70 / 10*